



TRAJECTORY OF A GLIDER

USING EULER'S METHOD IN PYTHON

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Summary

The kinematica of a glider can be described by system of differential equations. To solve these equations numerically, a computing program is needed. Besides the model should approximate the reality as close as possible and with so many variables involves, many simulations and fast solutions are needed. So computing program is unavoidable.

In this project a glider is modelled and its trajectory is solved numerically with Euler's method in Python computing language. From the results of the simulation can be concluded that Python computing language program is suitable to solve system of differential equation numerically.

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List of variables

$\mathbf{V}(\mathbf{t})$: The velocity of the glider at time $\mathbf{t}$ in $\frac{km}{h}$.

$\Theta(\mathbf{t})$: The angle of attack at time $\mathbf{t}$ in *gradians*.

$\mathbf{m}$: The mass of the glider in *kg*.

$\mathbf{C}_d$: The drag coefficient of the glider.

$\mathbf{g}$: The acceleration due to gravitational force $\frac{9.81*(60*60)^2}{1000}$
 $\frac{km}{h^2}$.

$\mathbf{C}_L$: The lift coefficient of the glider.

Chapter 1

Introduction

The force balances of a glider can be described as system of differential equations. To solve this equations numerically is a cumbersome works. Luckily we have computer to do this. Also the so many variables involve in the equations make the simulation with the computer program very suitable. So the glider can be modelled as close as can be.

In this project Euler's method is used to solve the differential equations and computing language used is Python.

In chapter 2 is presented that the main difference between a glider and an airplane is lack of engine. Further in the chapter is described what a glider is, how it takes off, glides and land.

In chapter 3 a glider is modelled. Further in the chapter is described; what variables are used, which stepsize is chosen, and some solutions in time of speed, angle of attack, position in x- and y-direction are plotted and in form of graphs are presented.

In chapter 4 some conclusions are drawn.

Chapter 2

Gliders: Airplanes Without Any Engines

While airplanes and gliders share many design, aerodynamic, and piloting factors, the lack of an engine fundamentally changes the way a glider flies.

2.1 Streamlined Fuselage

Since there is no engine taking up space, a glider is sized around the cargo it carries; the fuselage is designed to be as small and light as possible.

Most gliders have seats for two people in the small cockpit, with pilots sitting in a reclined position, versus powered airplanes, where pilots typically sit upright. Why the difference? By sitting reclined, the cockpit and canopy can be more streamlined, creating less drag in flight.

The surface of a glider's fuselage is designed to be as smooth as possible, allowing the plane to fly through the air with little parasitic drag. The earliest of gliders were constructed from wood covered with canvas; later versions were made from riveted structural aluminum skins. Unfortunately, the seams and rivets typical of aluminum significantly reduced performance due to parasitic drag, so gliders continued to adapt. Today, many advanced gliders are constructed from seamless materials like fiber glass and carbon fiber.



Figure 2.1: **Gliders constructed from fiber glass.** Allowing smoothest fuselage with little air drag.[3]

2.2 High Aspect Ratio Wings

Gliders have high aspect ratio wings, which means they are longer and narrower than wings on normal, powered airplanes.

Aspect ratio is calculated by dividing the square of the span of the wing by the area of the wing. As you can see in the diagram above, the Schleicher ASH 31 glider has an aspect of 33.5, while the Piper Cherokee has an aspect ratio of 5.6. High aspect ratio wings produced less induced drag, which is what makes them so efficient on gliders.

2.3 Control Surfaces

Like most airplanes, gliders use ailerons, a rudder, and an elevator to fly. Flaps are fitted on gliders to control descent rates by producing drag and increasing lift. Many modern gliders also use airbrakes or spoilers which, when used, drastically disrupt airflow over the wing, increasing drag and reducing lift.

Another significant difference between powered airplanes and gliders is that gliders normally have only one landing gear, situated directly below the pilot. Having only one gear saves a lot of weight, but what happens to the wings on takeoff and landing when you've got only one gear? The wingtips are pro-

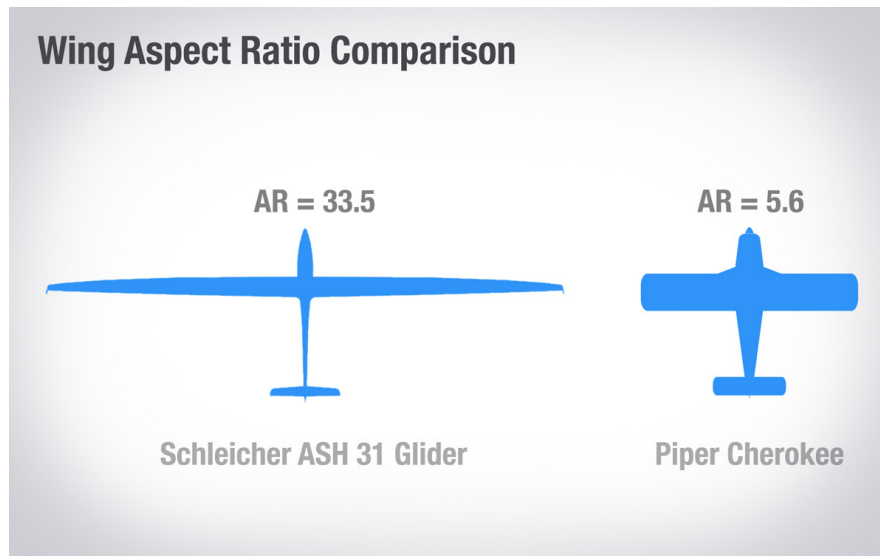


Figure 2.2: **Wing Aspect Ratio.** Gliders have high aspect ratio wings, which means the wings are longer and narrower. [3]

tected by skids or small wheels, and when the glider lands, it comes to rest on the main gear and one of the wingtips.

2.4 Glider Takeoff

Since they do not have engines, gliders normally use one of two methods to get off the ground:

2.4.1 Aero-Tow

A powered airplane tows the glider into the air using a long rope. Inside the cockpit, the glider pilot uses a quick-release mechanism to release the tow rope. Once the glider is at a desired altitude, the rope is released and the glider and tow plane turn in opposite directions.

2.4.2 Winch Launching

An engine on the ground powers a winch, connected to a cable launch system. The cable is then attached to the bottom of the glider. Once the winch is activated, the glider is pulled along the ground at high speed toward the winch and takes off. In a short amount of time, the glider gains substantial altitude

during this process and releases the winch line before continuing flight.

2.5 In Flight

Glide ratio measures the performance of an aircraft's glide; many modern gliders have a glide ratio better than 60:1. This means that if you start at an altitude of 1 kilometer, you can glide for 60 kilometers. In comparison, a Boeing 747 has a 15:1 glide ratio. But if glide ratio was the only thing keeping gliders in the air, they would not flying for very long. So how do they stay airborne? There are 3 main types of rising air that glider pilots use:

2.5.1 Thermals

These are columns of rising air created by the heating of the earth's surface. Air near the ground expands and rises as the surface of the earth is heated. Certain types of terrain absorb the sun more rapidly than others, like: asphalt parking lots, dark fields, rocky terrain, etc. These spots absorb heat, and heat the air above them, producing thermal air currents. Newly forming cumulus

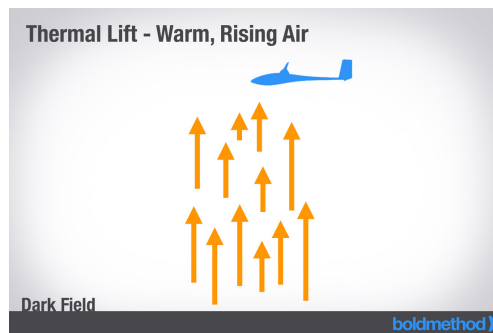


Figure 2.3: **Thermal.** Warm, rising air. [3]

clouds, or birds soaring without flapping their wings, are typically signs of thermal activity. When a glider pilot is "thermaling" they are finding and riding those thermal columns. And since the thermal can often cover only a small area, thermalling often involves a tight turn to stay inside the pocket of rising air.

2.5.2 Ridge Lift

This is created by winds blowing against mountains, hills or other ridges. Along the windward side of the mountain, a band of lift is formed where

air is redirected upward by the terrain. Typically, ridge lift extends only a few hundred feet higher than the terrain which produces it. Pilots have been known to go "ridge soaring" for thousands of kilometers along mountain chains.

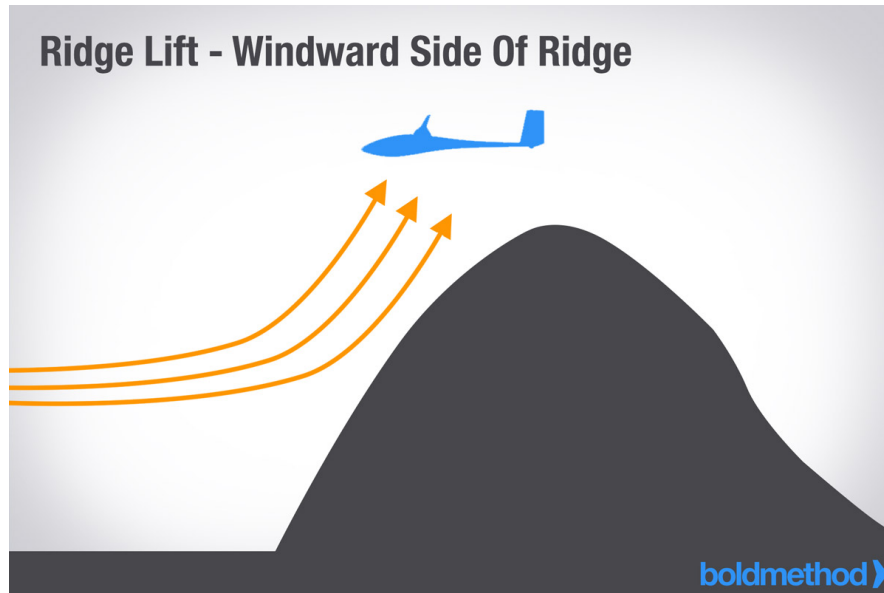


Figure 2.4: **Ridge Lift.** Windward side of ridge. [3]

2.5.3 Wave Lift

This is similar to ridge lift in that it is created when wind meets a mountain. However, wave lift is created on the leeward (downwind) side of the peaks by winds passing over top of the mountain. Wave lift can be identified by lenticular cloud formations - they look like flying saucers. Wave lift can reach thousands of meters high, and gliders riding on wave can reach altitudes of 10,000+ meters.

2.6 Ballast

Some gliders carry ballast tanks filled with water. Heavier gliders sink faster than lighter gliders. Glide ratio is not affected by weight because while a heavier glider may sink faster, it will do so at higher speed. The glider comes down faster with more weight, covering the same amount of distance; this is ideal for cross-country flying. A heavier glider, full of ballast, has a reduced climb rate and shorter flight endurance while in a lifting environment. Water

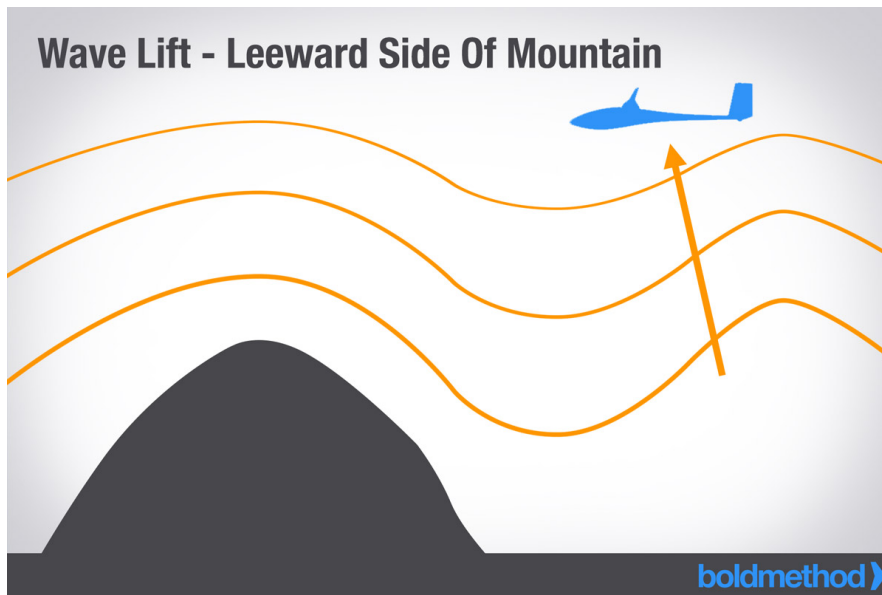


Figure 2.5: **Wave Lift.** Leeward side of mountain. [3]

ballast can be jettisoned at any time through dump valves to minimize these flight characteristics, and to slow down before landing.

2.7 Landing

Compared to landing in a powered airplane, there are a few key differences when flying a glider. First, gliders cannot add power if they will not make the touchdown zone. It may seem like a simple concept, but glider pilots are trained to judge their approach so they do not land short, and always wait until they are confident that they have the field made before introducing drag through flaps or spoilers.

The landing itself is not too different from that in any airplane, you flare until lift reduces, and try to touch down lightly. Since gliders have one wheel, it is a bit of a balancing act to keep the wings off the ground as long as possible.

Gliders are incredible aircraft, and with the right atmospheric conditions, can stay aloft for hours or days at a time. The careful aerodynamics design that goes into building one makes these birds swift and unique

Chapter 3

Trajectory of a Glider

3.1 Modelling the glider [1]

The glider flies under an angle θ with the horizon and with scalar velocity $\mathbf{V}$. The glider has no engine, and there are three forces working on the glider: gravity, drag and lift. In this simple model, both the drag and the lift are proportional to V^2 .

In aerodynamics the so called *angle of attack* is the angle between the wings and the oncoming airflow. We assume in this model that there is no wind and that the angle of attack is constant, so parameters $\mathbf{C_D}$ and $\mathbf{C_L}$ are assume to have constant values.



Figure 3.1: **Schematic glider model.** The forces working on a glider.

3.2 Differential equations of a glider [1]

- Forces balance in horizontal direction: $m \frac{dV}{dt} = -C_D * V^2 - m * g * \sin(\theta), \quad V(0) = 160,$

- Forces balance in vertical direction: $m * V * \frac{d\theta}{dt} = C_L * V^2 - m * g * \cos(\theta)$, $\theta(0) = 0$,

3.3 Chosing the variables

- Angle of attack θ . From [4], θ lies between -5° to 5°
- Initial velocity $V(0)$. In the begining for determining the right stepsizes initial velocity of 90 km/h and 160 km/h are chosen.
- Drag coeficient C_D and Lift coefficient C_L are constant. From [4], $C_D=0.008$ and $C_L=0.4$ are taken.

3.4 Determining the right stepsize

The result of the simulation using Euler's method in Python is recorded in following tables.

- $\theta(0) = 0; C_D=0.008; C_L=0.4$ and $V(0)=90$ km/h:

	dt	0.001	0.0001	0.00001
1.	V(10)	10.037	10.120	10.132
	error	-	83	12
2.	$\theta(10)$	0.09977	0.00805	0.00426
	error	-	-0.09172	-0.00379

Table 3.1: **Simulation result-1.** Stepsize $dt=0.00001$ give the least error.

- $\theta(0) = 0; C_D=0.008; C_L=0.4$ and $V(0)=160$ km/h:

	dt	0.001	0.0001	0.00001
1.	V(10)	10.031	10.119	10.132
	error	-	88	13
2.	$\theta(10)$	0.09933	0.00781	0.00406
	error	-	-0.09152	-0.00375

Table 3.2: **Simulation result-2.** Stepsize $dt=0.00001$ give the least error.

From the tables above the following variables are chosen: $\theta(0) = 0; C_D=0.008; C_L=0.4, V(0)=160$ km/h and $dt=0.00001$ to be used for further simulation (since these give the least error).

3.5 Solution for $V(t)$ and $\theta(t)$

The system differential equations from section 3.2 and the chosen variables from section 3.3 are calculated with Euler's method in Python. The following graphs are the result of the simulation:

- The solution of speed in time $V(t)$:
The velocity oscillates as seen in figure 3.2 hereunder. The glider speeds up and slows down, as at $t=0$ the the speed of the glider produces uplifting force which result in the rising of the angel of attack (the nose up). As the glider climbs up it decelerates due to gravity and the uplifting force decreases, the angel of attack begin decreasing (nose down), the glider goes down, pointing earth. Because of the gravity the glider begin to accelerate and thus speeding again which result in increasing uplifting force and angle of attack again. So the cycle begins anew.

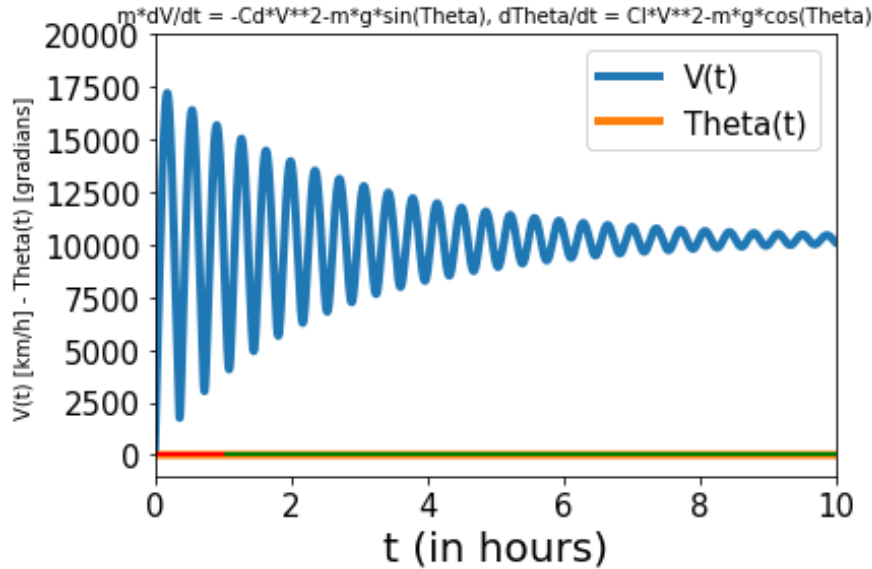


Figure 3.2: **Function of speed in time $V(t)$.** A bounded, oscillating and damped function.

- The solution of angle of attack in time $\theta(t)$. The solution is also a bounded, oscillating and damped function. The graph of the solution is displayed in Appendix A.

3.6 Solution for displacement in x-direction $x(t)$ and y-direction $y(t)$

The trajectory of the glider, trajectory through the sky is described by the horizontal position $x(t)$ and the vertical position, or height $y(t)$. The $x(t)$ and $y(t)$ are described by first-order differential equations of the following form: $\frac{dx}{dt} = f_x(V, \theta)$ and $\frac{dy}{dt} = f_y(V, \theta)$.

The solution for $V(t)$ and $\theta(t)$ can be approximated as follows:

$$V(t) = A + B * e^{-\mu * t} * \sin(\omega(t - \phi)) = 10135 + 7750 * e^{-0.1t} * \sin(16t)$$

$$\text{and for } \theta(t) = C + D * e^{-\mu * t} * \sin(\omega(t - \psi)) = 0.004 + 1.15 * e^{-0.1t} * \sin(17.7t)$$

With Euler's method in python both differential equations are calculated and the solutions are as follows:

1. Displacement in x-direction $x(t)$

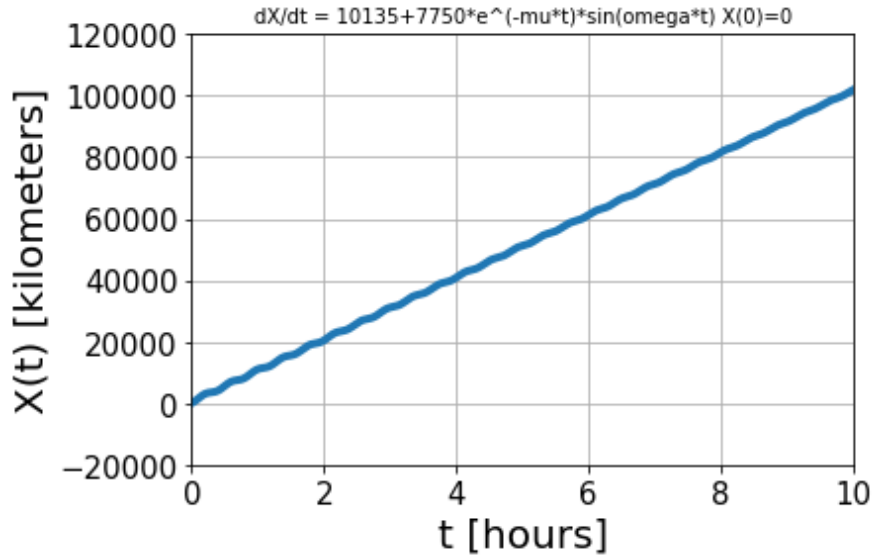


Figure 3.3: **X-position of the glider.** Begin by $x=0$ and travelled to the right.

2. Displacement in y-direction $y(t)$

From the graph can be seen that the glider makes up and down movements round oscillating height, as explained in chapter 3.5 the solutions for velocity and angle of attack. The displacements in this direction depends on the drag- and lift-coefficients (C_D and C_L) of the glider, which further depends on the material used and its wings geometry.

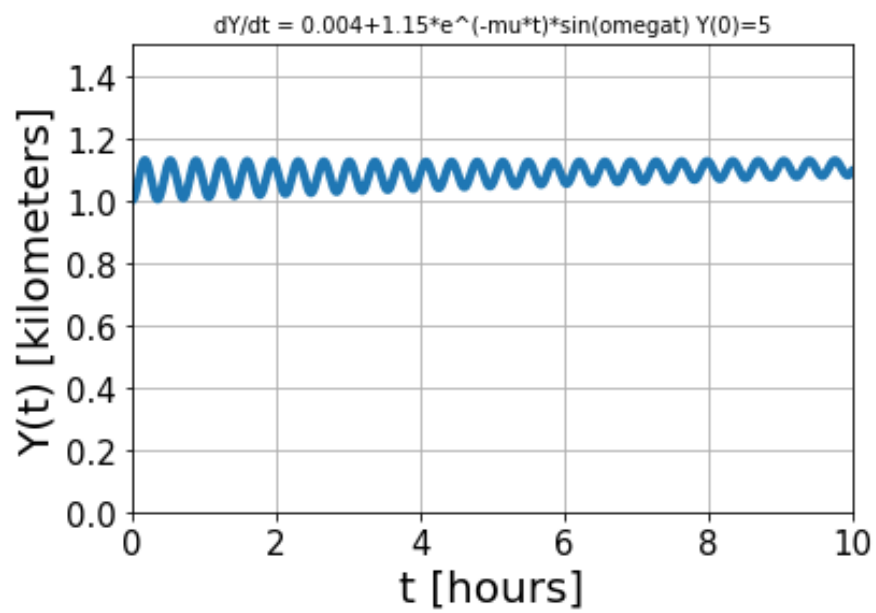


Figure 3.4: **Y-position of the glider.** Begin by $y=1$ km above the ground and oscillate around $y=1.1$ km.

Chapter 4

Conclusions

- The lack of engine in gliders makes them move differently than engined airplanes.
- In the Y-direction the glider makes up and down movements round oscillating height.
- The displacements in the Y-direction depends on the drag- and lift-coefficients (C_D and C_L) of the glider.
- The drag- and lift-coefficients (C_D and C_L) of the glider depends on the material used and its wings geometry.
- Python computing language program is suitable to calculate solution for system of differential equation with Euler's method.

Bibliography

- [1] MOOC team Mathematical Modelling Basics, Project; The trajectory of a glider, TUDelft, 01/05/2017.
- [2] Leslie Lamport, *LaTeX: a document preparation system*, Addison Wesley, Massachusetts, 2nd edition, 1994.
- [3] Martin Swayne, *How gliders fly, and how they are different than powered aircraft*, <http://www.boldmethod.com/blog/article>, 21/06/2017.
- [4] Thomas Hansen, *Modelling the performance of the standard cirrus glider using Navier-Stokes CFD*, <http://www.standardcirrus.org/Specifications.php>, 21/06/2017.

Appendix A

The graph of the solution of angle of attack in time $\theta(t)$

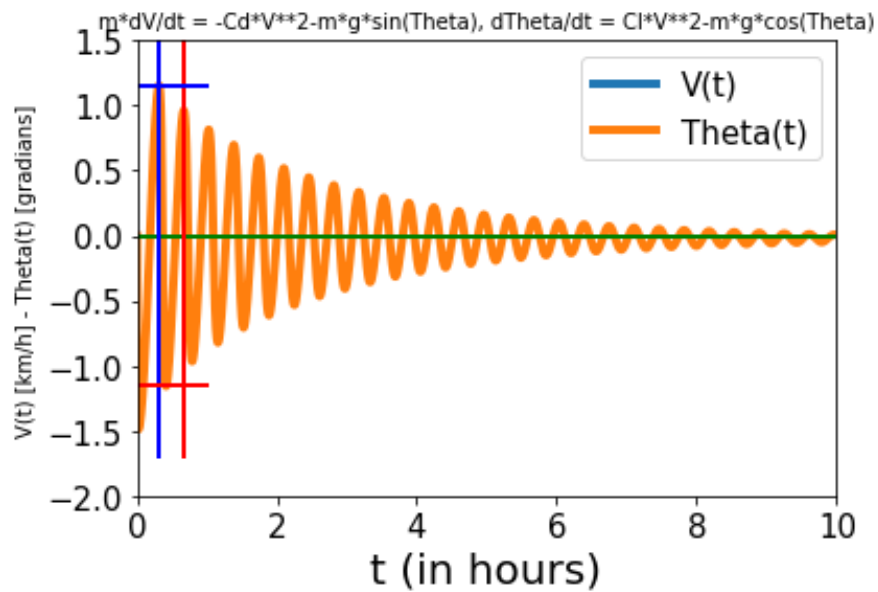


Figure A.1: **Function of angle of attack in time $\theta(t)$.** A bounded, oscillating and damped function.

Appendix B

Euler's method in Python

```

# Program : Euler's method for a system

# Author : MOOC team Mathematical Modelling Basics

# Created : May, 2017

import numpy as np

import matplotlib.pyplot as plt

from math import sin

from math import cos

print("Solution for  $m \frac{dV}{dt} = -C_d V^2 - m g \sin(\Theta)$ ,  $m \frac{d\Theta}{dt} = C_l V^2 - m g \cos(\Theta)$ ")

# Initializations

Dt = 0.00001 # timestep Delta t

V_init = 160 # initial velocity of V [km/hour]

Theta_init = (22/7)*0 # initial angle of attack of Theta=+-5 degrees

t_init = 0 # initial time

t_end = 10 # stopping time [hour]

m = 330 # gross weight [kg]

Cd = 0.008 # drag coefficient

g = (9.81*3600**2)/1000 # gravity [kg.km/hr**2]

Cl = 0.4

n_steps = int(round((t_end-t_init)/Dt)) # total number of timesteps

X = np.zeros(2) # create space for current X=[P,G]^T

```